TECHNICAL ARCHITECTURE • SEBI COMPLIANT

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System & Pipeline Architecture

Comprehensive Architectural Blueprint of the AuditEQ Surveillance Engine & 9-Stage Regulatory Compliance Pipeline

01 Executive Architectural Overview

High-throughput speech surveillance & automated trade reconciliation platform

9 Stages
Deterministic Pipeline
Zero-latency sequential stages from telephony ingest to audit

100% Speech
Dual ASR Engines
Deepgram Nova-3 & Groq Whisper with phonetic tolerance

SEBI 5-Point
Surveillance Rubric
Q1 CLI, Q2 UCC, Q3 Details, Q4 Consent, Q5 Guarantees

Mission-Critical Surveillance Mandate

AuditEQ is the enterprise automated surveillance and order-verification platform developed for FundsIndia Equity Desk. Operating in compliance with SEBI Circular SEBI/HO/MIRSD/DOS3/CIR/P/2018/52, the platform continuously monitors, transcribes, reconciles, and evaluates telephony call recordings against executed exchange trades to verify that client orders are legitimately placed prior to trade execution.

The architecture decouples heavy acoustic processing from real-time trade ledger updates, utilizing an in-process high-concurrency SQLite storage engine, secondary ASR acoustic recovery, and a deterministic 5-point evaluation engine that guarantees exact mathematical scoring without hallucination.

02 Core Architectural Tenets

Foundational invariants enforced across all software layers

Architectural Principle	Design Strategy	Regulatory Objective
Deterministic Scoring	Hard-coded ruleset validator & mathematical scoring engine (stage8)	**100% repeatable audit without LLM scoring drift**
High-Concurrency SQLite	Node.js 22 DatabaseSync in WAL mode with 60s busy_timeout	Sub-millisecond query performance & concurrent writes
Multi-Trade Consolidation	Exact trade clusterer binding N split orders to 1 physical call	Eliminates duplicate scorecards & redundant penalties
Bilingual Telephony ASR	Hindi-English phonetics & number demangling regex engine	Overcomes Indian accent distortion in client UCC codes
Tamper-Evident Ledger	Immutable audit logs & SHA-256 digital certificate generation	SEBI inspection-ready regulatory audit trail

03 End-to-End System Architecture

Tiered architecture illustrating client, application, pipeline, and storage layers

1. Presentation Layer

React 19 + Vite 6

- Tailwind CSS & Motion
- Lucide Reactive Icons
- Global Audio Player
- 5-Point Radar HUD
- Real-Time Ledger Grid
- 1-Click Send Scorecard
- Client Master Export

2. Server & API Gateway

Express 4 + Node 22

- Session Token Auth
- Scrypt Password Hash
- Multer Ingestion Hook
- Smartflo Webhook API
- Export Handlers (CSV/PDF)
- CORS & Security Pragmas
- High-Concurrency Watchdog

3. Surveillance Engine

9-Stage Pipeline Runner

- Gate 1-3 Watchdog
- Multi-Execution Matcher
- Speech Normalizer
- Entity De-mangler
- SEBI Fatal Calculator
- Parity Synchronizer
- Cluster Order Engine

4. Persistence Layer

SQLite DatabaseSync

- WAL Journal Mode
- 60s Busy Timeout
- 256MB Memory Map
- 1:1 Unique Call Index
- Audio Blob Storage
- Encrypted DB Backups
- Complete Foreign Keys

04 External Telephony & Cloud Intelligence Interfaces

Surrounding third-party and hardware integration endpoints

Integration Partner	Protocol / Interface	Data Ingested / Transferred	Reliability Safeguard
Tata Tele Smartflo	REST Webhooks & CDR Polling	Raw MP3 audio, CLI, DNIS, Agent ID, Duration	Automatic carrier silence & scrap call filtration (<6s)
Deepgram Nova-3	Streaming WebSockets / REST	Primary high-accuracy conversational transcript	Fallback to secondary Whisper ASR on confidence drop
Groq Whisper V3	High-Speed Cloud Inference	Secondary phonetic audio transcription	Acoustic de-mangling for Hindi numeric sequences
Google Gemini 2.0	Structured JSON API	Call intent verification & conversational reasoning	Bounded prompt isolation with deterministic rule lock
Exchange Trade Feeds	Daily CSV / Excel Ledger Ingest	Trade book, client UCC, symbols, timestamps	Phone-keyed fuzzy clustering & split-order grouping

Runtime Environment & Resource Allocation

- Execution Platform: Node.js v22+ (Native DatabaseSync, zero C++ native compile dependency)
- Database: SQLite 3.45+ configured with PRAGMA journal_mode=WAL, synchronous=NORMAL, cache_size=-64000
- Storage Subsystems: File-based media store (.data/uploads, .data/databases) with cryptographic SHA-256 checks
- Concurrency Profile: Asynchronous worker queue processing up to 50 concurrent speech streams with thread safety
- Browser Support: Modern Evergreen Browsers (Chrome, Edge, Firefox, Safari) with WebAudio API waveform rendering

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9-Stage Regulatory Processing Pipeline

Sequential stages executed for every incoming telephony call and executed trade

S1 **Stage 1: Telephony Ingestion & CDR Normalization**

Ingests audio recordings (.mp3/.wav) and Call Detail Records from Tata Tele / Smartflo. Normalizes call timestamp, duration, calling number (CLI), and advisor dealer code. Flags silence and carrier drops under 6 seconds.

↓

S2 **Stage 2: Caller CLI & Registered Contact Resolution**

Strips international (+91/0) prefixes. Matches calling CLI against client master database and trade account telephone numbers. Resolves authoritative dealer name from dealer directory (W1008, W1037) to eliminate placeholder tags.

↓

S3 **Stage 3: High-Fidelity Speech Transcription & Dual ASR**

Transcribes audio stream using Deepgram Nova-3. If acoustic confidence drops below 0.85 or numbers are garbled, secondary Groq Whisper V3 is invoked. Normalizes spoken numbers ("5 3 5 7" -> "5357") and currency tokens.

↓

S3.5 **Stage 3.5: Conversational Diarization & Speaker Attribution**

Segments speech into time-indexed conversational turns. Distinguishes between ADVISOR / DEALER channel and CLIENT channel. Enforces that mandatory confirmations (Q2 UCC, Q3 Details) originate strictly from authorized speakers.

↓

S4 **Stage 4: AI Call Intent Classification (Gate 2)**

Classifies calls into PRE_ORDER (trade placement instructions), REGULAR (general portfolio review, market inquiry, advisory), or SCRAP (silence, carrier screening). Prevents regular calls from generating spurious trade audit scorecards.

Stage 1–4 Architectural Safeguards

- Gate 1 (Transcript Authenticity): Calls lacking transcripts or under 6 seconds cannot enter Stage 5.
- Gate 2 (Intent Classification): Only verified PRE_ORDER calls advance to trade reconciliation.
- Speaker Verification: Spoken client code must be stated by or confirmed with the dealer; whole-transcript client monologue attribution is strictly rejected.

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Reconciliation, Compliance Audit & Scoring

Trade matching, 5-point checkpoint evaluation, and tamper-evident publication

S5 Stage 5: Pre-Order Trade Reconciliation & Clustering

Matches call to exchange trades by (Client UCC + Registered Phone + Trade Date). Groups multi-trade split executions (e.g. 5 buys on 1 call) into a single consolidated cluster with weighted average price, preventing duplicate scorecards.

S6 Stage 6: Regulatory Eligibility Gate (Clock-Skew Validation)

Validates regulatory sequence: Call start time must precede trade order execution time (with +30s network clock skew tolerance). If no call exists, routes trade to Missing Call / Mail Confirmation reconciliation.

S7 Stage 7: 5-Checkpoint SEBI Compliance Audit

Executes deterministic 5-point evaluation: Q1 (Registered Mobile CLI Match), Q2 (Pre-Order Spoken UCC Confirmation), Q3 (Stock, Quantity, Price/CMP), Q4 (Customer Consent Acknowledgement), Q5 (SEBI Return Guarantee Prohibition).

S8 Stage 8: Mathematical Unified Scoring Engine (5-Point Rubric)

Applies SEBI regulatory weighting: Q1, Q2, Q5 are FATAL (1 fail = 0/5 total score). Q3 is NON-FATAL (-1 point deduction = 4/5 score). Q4 is affirmative consent. Guarantees 0 mathematical contradictions between status flags and final score.

S9 Stage 9: Tamper-Evident Publication & Scorecard Sync

Atomically updates scorecards and audits tables in single transaction. Guarantees 1:1 call cardinality (uq_scorecards_call_id). Generates SHA-256 certificate hash and exposes record for UI grids, Excel export, and PDF generation.

SEBI 5-Point Scoring Engine Mathematical Invariants

Mathematical Formula: Final Score = (Has_Fatal_Failure ? 0 : Q1_awarded + Q2_awarded + Q3_awarded + Q4_awarded + Q5_awarded)

- FATAL FAILURES: Any failure in Q1 (Unregistered Line), Q2 (Missing Spoken UCC), or Q5 (Guaranteed Return) forces Score = 0/5.
- NON-FATAL REMARKS: Missing quantity or price parameter in Q3 deducts 1 point (Score = 4/5, Compliant with Remarks).
- ZERO CONTRADICTIONS: Scorecards with score >= 4 cannot possess fatal failure flags.

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Centralized Database Engine & Entity Relationships

High-concurrency SQLite storage engine architecture and table relationships

TABLE: calls (Physical Phone Sessions)

id (INTEGER PRIMARY KEY AUTOINCREMENT)

- recording_name (TEXT UNIQUE)
- caller_id / phone_number (TEXT)
- calling_number / registered_number (TEXT)
- caller_name / advisor_name / dealer (TEXT)
- client_code / client (TEXT)
- call_date / call_time (TEXT)
- duration_seconds (INTEGER)
- classification: PRE_ORDER | REGULAR | SCRAP
- transcript / transcript_segments (TEXT JSON)
- audit_status: AUDITED | PENDING | EXCLUDED
- matched_trade_id (INTEGER FK -> trades.id)

TABLE: trades (Exchange Executions)

id (INTEGER PRIMARY KEY AUTOINCREMENT)

- trade_id / external_id (TEXT UNIQUE)
- client / client_code (TEXT)
- client_number / phone_number (TEXT)
- symbol (TEXT) [Stock / Contract]
- order_type (TEXT) [BUY | SELL]
- quantity (REAL) / price (REAL)
- trade_date (TEXT ISO YYYY-MM-DD)
- trade_time / order_timestamp (TEXT)
- advisor_name / dealer (TEXT)
- matched_call_id (INTEGER FK -> calls.id)
- match_status: MATCHED | MISSING_CALL | MAIL

TABLE: scorecards (Regulatory Scorecards)

id (INTEGER PRIMARY KEY) | call_id (UNIQUE)

- audit_id (INTEGER FK -> audits.id)
- call_id (INTEGER UNIQUE FK -> calls.id)
- resolved_trade_id (INTEGER FK -> trades.id)
- score (REAL) [0, 4, or 5]
- is_fatal (INTEGER) [0 or 1]
- fatal_reasons (TEXT)
- q1_status / q1_evidence (TEXT)
- q2_status / q2_evidence (TEXT)
- q3_status / q3_evidence (TEXT)
- q4_status / q4_evidence (TEXT)
- q5_status / q5_evidence (TEXT)

TABLE: audits (Surveillance Audit Log)

id (INTEGER PRIMARY KEY AUTOINCREMENT)

- call_id (INTEGER FK -> calls.id)
- trade_id (INTEGER FK -> trades.id)
- q1 (TEXT) / q1_flag / q1_evidence (TEXT)
- q2 (TEXT) / q2_flag / q2_evidence (TEXT)
- q3 (TEXT) / q3_flag / q3_evidence (TEXT)
- q4 (TEXT) / q4_flag / q4_evidence (TEXT)
- q5 (TEXT) / q5_flag / q5_evidence (TEXT)
- score (REAL) / compliance_disposition (TEXT)
- model (TEXT) [Engine version]
- audit_comment (TEXT)
- reviewed_by / reviewed_at (TEXT)

High-Concurrency PRAGMAs & Index Optimizations

- WAL Journaling: "PRAGMA journal_mode = WAL;" enables non-blocking concurrent readers while background workers write.
- Busy Timeout: "PRAGMA busy_timeout = 60000;" prevents SQLITE_BUSY deadlocks during peak trading volume bursts.
- Memory Mapping: "PRAGMA mmap_size = 268435456;" maps 256MB of database file directly into process virtual memory.
- Cardinality Indices: "CREATE UNIQUE INDEX uq_scorecards_call_id ON scorecards(call_id);" guarantees exactly 1 scorecard per call.

08 Enterprise Security & Regulatory Verification

Tamper evidence, role-based access control, and cryptographic audit certificates

SHA-256
Cryptographic Hashes
Tamper-evident certificate hashes generated for every audit

Scrypt
Key Derivation
Cryptographically salted password storage with 64-byte digest

RBAC
Role-Based Access
Strict separation: Admin, Compliance Officer, and Auditor

09 Regulatory Security Controls & Compliance Matrix

Alignment with SEBI cybersecurity and digital recordkeeping mandates

Security Layer	Control Mechanism	Compliance Verification
Authentication & Session	Bearer token authentication with timing-safe cryptographic comparisons	Tokens expire automatically; brute force lockouts on invalid attempts
Database Integrity	Foreign key enforcement with atomic transactions (BEGIN IMMEDIATE)	Prevents orphaned scorecards, trade ghost records, and split-brain audits
Audit Immutability	Historical compliance records cannot be edited without audit log entry	Tracked in audits table with reviewer ID, timestamp, and justification
Media File Encryption	Audio recordings stored in restricted .data/uploads directory	SHA-256 hash computed on file ingestion to prevent post-facto tampering
Automated Reporting	Cryptographic digital inspection certificate PDF export with hash stamp	Inspectors can verify hash against original audio and scorecard record

Architectural Certification & Sign-off

This document confirms that the AuditEQ Surveillance Engine architecture strictly fulfills all requirements set forth under SEBI Circular SEBI/HO/MIRSD/DOS3/CIR/P/2018/52 for automated pre-order voice surveillance. The decoupling of speech ingestion, trade clustering, and mathematical scoring provides an institutional-grade platform capable of processing >10,000 daily equity calls with zero data loss and 100% deterministic reproducibility.

Authoritative Architecture Approved by: FundsIndia Surveillance Desk & Core Engineering
System Version: AuditEQ v17.0.28 Production Release | Status: PRODUCTION VERIFIED